

```
CREATE DATABASE IF NOT EXISTS `smart_laundry`  
  CHARACTER SET utf8mb4  
  COLLATE utf8mb4_unicode_ci;
```

```
USE `smart_laundry`;
```

```
CREATE TABLE IF NOT EXISTS users (  
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
  student_code VARCHAR(20) NULL UNIQUE,  
  email VARCHAR(255) NOT NULL UNIQUE,  
  password_hash VARCHAR(255) NOT NULL,  
  full_name VARCHAR(150) NOT NULL,  
  phone VARCHAR(20) NULL,
```

```
  role ENUM(  
    'USER',  
    'DORM_ADMIN',  
    'SYSTEM_ADMIN'  
  ) NOT NULL DEFAULT 'USER',
```

```
  status ENUM(  
    'ACTIVE',  
    'SUSPENDED',  
    'PENDING'  
  ) NOT NULL DEFAULT 'ACTIVE',
```

```
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP  
    ON UPDATE CURRENT_TIMESTAMP,
```

```
  INDEX idx_users_role_status (role, status)  
) ENGINE=InnoDB;
```

```
CREATE TABLE IF NOT EXISTS dormitories (  
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,  
  dorm_name VARCHAR(150) NOT NULL,  
  address TEXT NULL,  
  contact_phone VARCHAR(20) NULL,
```

```
  status ENUM(  
    'ACTIVE',  
    'INACTIVE'  
  ) NOT NULL DEFAULT 'ACTIVE',
```

```
  created_by BIGINT UNSIGNED NOT NULL,
```

```
created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP  
ON UPDATE CURRENT_TIMESTAMP,
```

```
UNIQUE KEY uq_dormitory_name (dorm_name),
```

```
CONSTRAINT fk_dormitory_creator  
  FOREIGN KEY (created_by)  
  REFERENCES users(id)  
) ENGINE=InnoDB;
```

```
CREATE TABLE IF NOT EXISTS rooms (  
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
```

```
  dormitory_id BIGINT UNSIGNED NOT NULL,
```

```
  room_number VARCHAR(20) NOT NULL,  
  floor_number INT NOT NULL,
```

```
  room_status ENUM(  
    'AVAILABLE',  
    'OCCUPIED',  
    'MAINTENANCE'  
  ) NOT NULL DEFAULT 'OCCUPIED',
```

```
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP  
  ON UPDATE CURRENT_TIMESTAMP,
```

```
  UNIQUE KEY uq_room_dormitory_number  
    (dormitory_id, room_number),
```

```
  CONSTRAINT fk_rooms_dormitory  
    FOREIGN KEY (dormitory_id)  
    REFERENCES dormitories(id)  
    ON DELETE CASCADE  
) ENGINE=InnoDB;
```

```
CREATE TABLE IF NOT EXISTS room_members (  
  room_id BIGINT UNSIGNED NOT NULL,  
  user_id BIGINT UNSIGNED NOT NULL,
```

```
  member_role ENUM(  
    'OWNER',  
    'RESIDENT'  
  ) NOT NULL DEFAULT 'RESIDENT',
```

```

joined_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

PRIMARY KEY (room_id, user_id),

CONSTRAINT fk_room_member_room
  FOREIGN KEY (room_id)
  REFERENCES rooms(id)
  ON DELETE CASCADE,

CONSTRAINT fk_room_member_user
  FOREIGN KEY (user_id)
  REFERENCES users(id)
  ON DELETE CASCADE
) ENGINE=InnoDB;

CREATE TABLE IF NOT EXISTS washing_machines (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,

  dormitory_id BIGINT UNSIGNED NOT NULL,

  machine_code VARCHAR(30) NOT NULL UNIQUE,
  machine_name VARCHAR(100) NOT NULL,
  location VARCHAR(150) NOT NULL,

  machine_type ENUM(
    'WASHING_MACHINE',
    'DRYER'
  ) NOT NULL DEFAULT 'WASHING_MACHINE',

  status ENUM(
    'AVAILABLE',
    'RESERVED',
    'RUNNING',
    'MAINTENANCE',
    'OUT_OF_SERVICE'
  ) NOT NULL DEFAULT 'AVAILABLE',

  price_per_cycle DECIMAL(10,2) NOT NULL DEFAULT 0.00,

  estimated_minutes INT NOT NULL DEFAULT 60,

  installed_at DATE NULL,

  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
  ON UPDATE CURRENT_TIMESTAMP,

```

```
INDEX idx_machine_dormitory_status  
(dormitory_id, status),
```

```
CONSTRAINT fk_machine_dormitory  
  FOREIGN KEY (dormitory_id)  
  REFERENCES dormitories(id)  
  ON DELETE CASCADE  
) ENGINE=InnoDB;
```

```
CREATE TABLE IF NOT EXISTS bookings (  
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
```

```
  machine_id BIGINT UNSIGNED NOT NULL,  
  user_id BIGINT UNSIGNED NOT NULL,  
  room_id BIGINT UNSIGNED NULL,
```

```
  booking_code VARCHAR(50) NOT NULL UNIQUE,
```

```
  booking_date DATE NOT NULL,  
  start_time TIME NOT NULL,  
  end_time TIME NOT NULL,
```

```
  total_price DECIMAL(10,2) NOT NULL DEFAULT 0.00,
```

```
  status ENUM(  
    'PENDING',  
    'CONFIRMED',  
    'RUNNING',  
    'COMPLETED',  
    'CANCELLED',  
    'NO_SHOW'  
  ) NOT NULL DEFAULT 'PENDING',
```

```
  cancelled_reason VARCHAR(255) NULL,
```

```
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP  
  ON UPDATE CURRENT_TIMESTAMP,
```

```
INDEX idx_booking_machine_date  
(machine_id, booking_date),
```

```
INDEX idx_booking_user_date  
(user_id, booking_date),
```

```
INDEX idx_booking_status
```

(status),

CONSTRAINT fk_booking_machine
FOREIGN KEY (machine_id)
REFERENCES washing_machines(id)
ON DELETE RESTRICT,

CONSTRAINT fk_booking_user
FOREIGN KEY (user_id)
REFERENCES users(id)
ON DELETE RESTRICT,

CONSTRAINT fk_booking_room
FOREIGN KEY (room_id)
REFERENCES rooms(id)
ON DELETE SET NULL
) ENGINE=InnoDB;

CREATE TABLE IF NOT EXISTS payments (
id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,

booking_id BIGINT UNSIGNED NOT NULL,

amount DECIMAL(10,2) NOT NULL,

payment_method ENUM(
'CASH',
'PROMPTPAY',
'BANK_TRANSFER'
) NOT NULL,

payment_status ENUM(
'PENDING',
'PAID',
'FAILED',
'REFUNDED'
) NOT NULL DEFAULT 'PENDING',

transaction_reference VARCHAR(100) NULL,

paid_at DATETIME NULL,

created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

UNIQUE KEY uq_payment_booking (booking_id),

CONSTRAINT fk_payment_booking

```
    FOREIGN KEY (booking_id)
    REFERENCES bookings(id)
    ON DELETE CASCADE
) ENGINE=InnoDB;
```

```
CREATE TABLE IF NOT EXISTS machine_maintenance (
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
```

```
    machine_id BIGINT UNSIGNED NOT NULL,
    reported_by BIGINT UNSIGNED NULL,
```

```
    problem_type ENUM(
        'NOT_WORKING',
        'WATER_PROBLEM',
        'ELECTRICAL_PROBLEM',
        'DOOR_PROBLEM',
        'SPIN_PROBLEM',
        'OTHER'
    ) NOT NULL DEFAULT 'OTHER',
```

```
    problem_title VARCHAR(255) NOT NULL,
    description TEXT NULL,
```

```
    status ENUM(
        'PENDING',
        'REPAIRING',
        'COMPLETED',
        'CANCELLED'
    ) NOT NULL DEFAULT 'PENDING',
```

```
    reported_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    started_repair_at DATETIME NULL,
    completed_at DATETIME NULL,
```

```
    technician_name VARCHAR(150) NULL,
    repair_cost DECIMAL(10,2) NULL,
```

```
    INDEX idx_maintenance_machine_status
    (machine_id, status),
```

```
    CONSTRAINT fk_maintenance_machine
    FOREIGN KEY (machine_id)
    REFERENCES washing_machines(id)
    ON DELETE CASCADE,
```

```
    CONSTRAINT fk_maintenance_reporter
    FOREIGN KEY (reported_by)
```

```
REFERENCES users(id)
ON DELETE SET NULL
) ENGINE=InnoDB;
```

```
CREATE TABLE IF NOT EXISTS machine_usage_logs (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,

  machine_id BIGINT UNSIGNED NOT NULL,
  booking_id BIGINT UNSIGNED NULL,
  user_id BIGINT UNSIGNED NOT NULL,

  started_at DATETIME NOT NULL,
  finished_at DATETIME NULL,

  usage_minutes INT NULL,

  usage_status ENUM(
    'STARTED',
    'COMPLETED',
    'INTERRUPTED'
  ) NOT NULL DEFAULT 'STARTED',

  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

  INDEX idx_usage_machine
    (machine_id, started_at),

  INDEX idx_usage_user
    (user_id, started_at),

  CONSTRAINT fk_usage_machine
    FOREIGN KEY (machine_id)
    REFERENCES washing_machines(id)
    ON DELETE CASCADE,

  CONSTRAINT fk_usage_booking
    FOREIGN KEY (booking_id)
    REFERENCES bookings(id)
    ON DELETE SET NULL,

  CONSTRAINT fk_usage_user
    FOREIGN KEY (user_id)
    REFERENCES users(id)
    ON DELETE CASCADE
) ENGINE=InnoDB;
```

```

CREATE TABLE IF NOT EXISTS machine_status_logs (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,

  machine_id BIGINT UNSIGNED NOT NULL,

  old_status VARCHAR(50) NULL,
  new_status VARCHAR(50) NOT NULL,

  changed_by BIGINT UNSIGNED NULL,

  reason VARCHAR(255) NULL,

  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

  INDEX idx_machine_status_log
    (machine_id, created_at),

  CONSTRAINT fk_status_log_machine
    FOREIGN KEY (machine_id)
    REFERENCES washing_machines(id)
    ON DELETE CASCADE,

  CONSTRAINT fk_status_log_user
    FOREIGN KEY (changed_by)
    REFERENCES users(id)
    ON DELETE SET NULL
) ENGINE=InnoDB;

```

```

CREATE TABLE IF NOT EXISTS notifications (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,

  user_id BIGINT UNSIGNED NOT NULL,
  booking_id BIGINT UNSIGNED NULL,
  machine_id BIGINT UNSIGNED NULL,

  notification_type ENUM(
    'BOOKING',
    'REMINDER',
    'MACHINE',
    'MAINTENANCE',
    'PAYMENT',
    'SYSTEM'
  ) NOT NULL,

  title VARCHAR(255) NOT NULL,
  message TEXT NOT NULL,

```

```
is_read BOOLEAN NOT NULL DEFAULT FALSE,

created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

INDEX idx_notifications_user
  (user_id, is_read, created_at),

CONSTRAINT fk_notification_user
  FOREIGN KEY (user_id)
  REFERENCES users(id)
  ON DELETE CASCADE,

CONSTRAINT fk_notification_booking
  FOREIGN KEY (booking_id)
  REFERENCES bookings(id)
  ON DELETE SET NULL,

CONSTRAINT fk_notification_machine
  FOREIGN KEY (machine_id)
  REFERENCES washing_machines(id)
  ON DELETE SET NULL
) ENGINE=InnoDB;
```

```
CREATE TABLE IF NOT EXISTS machine_reviews (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,

  machine_id BIGINT UNSIGNED NOT NULL,
  booking_id BIGINT UNSIGNED NULL,
  user_id BIGINT UNSIGNED NOT NULL,

  rating TINYINT UNSIGNED NOT NULL,

  comment TEXT NULL,

  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

  UNIQUE KEY uq_review_booking_user
    (booking_id, user_id),

  CONSTRAINT chk_rating
    CHECK (rating BETWEEN 1 AND 5),

  CONSTRAINT fk_review_machine
    FOREIGN KEY (machine_id)
    REFERENCES washing_machines(id)
    ON DELETE CASCADE,
```

```

CONSTRAINT fk_review_booking
  FOREIGN KEY (booking_id)
  REFERENCES bookings(id)
  ON DELETE SET NULL,

CONSTRAINT fk_review_user
  FOREIGN KEY (user_id)
  REFERENCES users(id)
  ON DELETE CASCADE
) ENGINE=InnoDB;

CREATE TABLE IF NOT EXISTS audit_logs (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,

  actor_user_id BIGINT UNSIGNED NULL,

  action VARCHAR(120) NOT NULL,

  entity_type VARCHAR(80) NULL,

  entity_id BIGINT UNSIGNED NULL,

  details JSON NULL,

  ip_address VARCHAR(64) NULL,

  result ENUM(
    'SUCCESS',
    'FAILURE'
  ) NOT NULL DEFAULT 'SUCCESS',

  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

  INDEX idx_audit_created_at (created_at),
  INDEX idx_audit_actor (actor_user_id),

  CONSTRAINT fk_audit_actor
    FOREIGN KEY (actor_user_id)
    REFERENCES users(id)
    ON DELETE SET NULL
) ENGINE=InnoDB;

```